

Dhairya Joshi

B.Tech., Indian Institute of Technology (IIT), Jodhpur (Minor in AI)
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EDUCATION

B.Tech – Indian Institute of Technology Jodhpur – CGPA: 8.43 2024 – 2028
Senior Secondary – Percentage: 92.2% 2024
Secondary – Percentage: 96.4% 2022

EXPERIENCE

Undergraduate Research Assistant – FinSense AI GitHub **Dec. 2025 – Jan. 2026**
Supervisor: Prof. G. Venkat Ram Reddy, SME, IIT Jodhpur

- Architected 80+ leakage-safe features from market, sentiment, and Google Trends data across 20+ stocks.
- Cut next-day return prediction error 20% via a Random Forest–Gradient Boosting–XGBoost ensemble.
- Deployed a FastAPI backend exposing REST APIs to a live frontend with real-time progress tracking, serving predictions across the full tracked ticker universe alongside an event-driven backtesting engine (fixed/trailing/ATR stop-loss) evaluated via Sharpe ratio, max drawdown, and win-rate.
- Diagnosed and resolved production JSON serialization failures from NaN/Inf propagation in external financial APIs, eliminating recurring backend crashes.

PROJECTS

CheckerPlane AI – Low-Latency LLM Guardrail Proxy GitHub **July 2026 – Aug. 2026**
Rust, ONNX Runtime, C++, DeBERTa, BERT, Tokio, Async Concurrency

- Architected a multi-tier AI safety proxy in pure Rust with ONNX Runtime, replacing sequential Python guardrail chains with parallel async execution, cutting median guardrail latency from ~550ms to ~67ms (~8x reduction).
- Implemented a 4-gate parallel input rail (toxicity, PII/Aadhaar NER, coherence, prompt-injection) with early-exit cancellation and a <0.1ms Tier-0 heuristic screen, filtering unsafe requests before deeper inference.
- Trained a DeBERTa cross-encoder NLI hallucination gate scoring output grounding pre-delivery, achieving ~88% average gate accuracy with zero GPU dependency via quantised ONNX models.

Delta Hedging & Adversarial Stock Path Generation Simulator GitHub **May 2026 – June 2026**
Python, TensorFlow, NumPy, Pandas, Monte Carlo Simulation, Black-Scholes Model

- Built a quantitative finance simulator benchmarking Black-Scholes delta hedging against deep learning-based hedging strategies across 10,000+ synthetic market scenarios generated through Monte Carlo simulation.
- Architected and trained a GRU-based deep hedging model using a custom Conditional Value-at-Risk (CVaR) loss function, reducing CVaR_{5%} tail risk from Rs. 500 (Black-Scholes baseline) to Rs. 290, a 42% improvement.
- Designed an adversarial training framework where a recurrent generator learns worst-case market conditions, further cutting CVaR_{5%} to Rs. 100 - an 80% drop versus the Black-Scholes baseline.
- Processed 8,500+ minute-level NIFTY options data points, computing implied volatility and Black-Scholes Greeks for model calibration and evaluation.

LLM Response Preference Classification – Kaggle Competition GitHub **Aug. 2026 – Sept. 2026**
Python, PyTorch, HuggingFace Transformers, Llama 3, Gemma 2, xFormers, Multi-GPU Inference

- Fine-tuned Llama-3 and Gemma-2 sequence-classification models on ~57,400 datapoints to predict human preference between pairs of LLM chatbot responses, framing the task as a 3-class (win A / win B / tie) classification problem.
- Engineered a pipelined multi-GPU inference system splitting each Llama-3 model across two GPUs, using variable-length attention (xFormers block-diagonal masking) and sharded max-token batching to eliminate padding waste and maximize throughput.
- Benchmarked the Llama-3/Gemma-2 ensemble against an LSTM baseline, reducing log-loss from 1.13208 to 0.83039 (~27% improvement), motivating the switch to transformer-based sequence classification.
- Ensembled Llama-3 and Gemma-2 predictions with test-time augmentation, achieving a 0.83039 log-loss - a top-1% finish among 224 entries.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Rust

Machine Learning & Deep Learning: Scikit-Learn, Random Forest, XGBoost, Gradient Boosting, Feature Engineering, PyTorch, TensorFlow, Keras, CNN, RNN, LSTM, GRU, GANs, Transformer Architectures (Llama, Gemma, BERT, DeBERTa)

Generative AI / LLMs & Systems: LLM Fine-Tuning, RAG, LangChain, Vector Databases, Prompt Engineering, LLM Guardrails & Safety Pipelines, FastAPI, REST APIs, ONNX Runtime, Async Concurrency, Multi-GPU Pipelined Inference

Data Science: NumPy, Pandas, Matplotlib

Tools: Git, GitHub, Jupyter Notebook, VS Code, LaTeX

ACHIEVEMENTS

- Secured AIR 3426 among 1.5M+ candidates in JEE Main, achieving 99.9 percentile in Mathematics and ranking among the top 4% candidates in JEE Advanced.
- Ranked 2nd of 224 teams (top 1%) in the Kaggle ‘LLM Classification Finetuning’ competition, competing solo.

RELEVANT COURSEWORK

Data Structures & Algorithms (A) | Machine Learning (A-) | Scientific Computations (A-) | Deep Learning | Probability & Statistics | Natural Language Understanding (NLU) | Mathematics-I | Mathematics-II